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PRACTICAL 1

A) AIM: A client server-based program using TCP to find if the number entered is prime.

CODE:

1. tcpServerPrime.java :

```
import java.net.*;
import java.io.*;

class tcpServerPrime {
   public static void main(String args[]) {
      try {
         ServerSocket ss = new ServerSocket(8001);
         System.out.println("Server Started..... ");

         Socket s = ss.accept();

         DataInputStream in =
               new DataInputStream(s.getInputStream());

         int x = in.readInt();

         DataOutputStream otc =
               new DataOutputStream(s.getOutputStream());

         int y = x / 2;
         boolean prime = true;

         if (x <= 1) {
            prime = false;
         } else {
            for (int i = 2; i <= y; i++) {
               if (x % i == 0) {
```

```
                    prime = false;
                    break;
                }
            }
        }

        if (prime) {
            otc.writeUTF(x + " is Prime");
        } else {
            otc.writeUTF(x + " is not Prime");
        }

        in.close();
        otc.close();
        s.close();
        ss.close();
      } catch (Exception e) {
        System.out.println(e.toString());
      }
    }
  }
```

2. tcpClientPrime.java :

```
import java.net.*;
import java.io.*;

class tcpClientPrime {
  public static void main(String args[]) {
    try {
      Socket cs = new Socket("localhost", 8001);

      BufferedReader infu = new BufferedReader(
```

```
            new InputStreamReader(System.in));

        System.out.println("Enter a number : ");
        int a = Integer.parseInt(infu.readLine());

        DataOutputStream out =
            new DataOutputStream(cs.getOutputStream());

        out.writeInt(a);

        DataInputStream in =
            new DataInputStream(cs.getInputStream());

        System.out.println(in.readUTF());

        in.close();
        out.close();
        cs.close();
      } catch (Exception e) {
        System.out.println(e.toString());
      }
   }
}
```

Output:

```
D:\M.SC.IT> javac tcpServerPrime.java
D:\M.SC.IT>java tcpServerPrime
Server Started.....
D:\M.SC.IT>javac tcpServerPrime.java
D:\M.SC.IT>java tcpServerPrime
Server Started.....
```

```
D:\M.SC.IT>javac tcpClientPrime.java

D:\M.SC.IT>java tcpClientPrime

Enter a number:

7

7 is Prime

D:\M.SC.IT>javac tcpClientPrime.java

D:\M.SC.IT>java tcpClientPrime

Enter a number :

8

8 is not Prime
```

B) AIM : A client server TCP based chatting application.

1. ChatServer.java

```
import  java.net.*;
import java.io.*;

class ChatServer {
   public static void main(String args[]) {
      try {
         ServerSocket ss = new ServerSocket(8000);
         System.out.println("Waiting for client to connect..");

         Socket s = ss.accept();

         BufferedReader br = new BufferedReader(
               new InputStreamReader(System.in));

         DataOutputStream out =
               new DataOutputStream(s.getOutputStream());

         DataInputStream in =
               new DataInputStream(s.getInputStream());

         String receive, send;

         while ((receive = in.readLine()) != null) {
            if (receive.equals("STOP"))
               break;

            System.out.println("Client Says : " + receive);
            System.out.print("Server Says : ");
```

```
            send = br.readLine();
            out.writeBytes(send + "\n");

            if (send.equals("STOP"))
                break;
        }

        br.close();
        in.close();
        out.close();
        s.close();
        ss.close();
      } catch (Exception e) {
        e.printStackTrace();
      }
   }
}
```

2. ChatClient.java

```
import java.net.*;
import java.io.*;

class ChatClient {
   public static void main(String args[]) {
      try {
        Socket s = new Socket("localhost", 8000);

        BufferedReader br = new BufferedReader(
              new InputStreamReader(System.in));

        DataOutputStream out =
              new DataOutputStream(s.getOutputStream());
```

```
            DataInputStream in =
                    new DataInputStream(s.getInputStream());

            String msg;

            System.out.println("To stop chatting with server type STOP");
            System.out.print("Client Says: ");

            while ((msg = br.readLine()) != null) {
               out.writeBytes(msg + "\n");

               if (msg.equals("STOP"))
                  break;

               System.out.println("Server Says : " + in.readLine());
               System.out.print("Client Says : ");
            }

            br.close();
            in.close();
            out.close();
            s.close();
         } catch (Exception e) {
            e.printStackTrace();
         }
      }
   }
```

Output:

Server:

```
E:\Ds_Yugi>java ChatServer
Waiting for client to connect..
Client Says : Hii...
Server Says : Hello
Client Says : How are you?
Server Says : Fine, thank you..
```

Client:

```
E:\Ds_Yugi>java ChatClient
To stop chatting with server type STOP
Client Says: Hii...
Server Says : Hello
Client Says : How are you?
Server Says : Fine, thank you..
Client Says : STOP
```

Practical 2

A) AIM : A client server-based program using UDP to find if the number entered is even or odd.

CODE :

1. udpServerEO.java

```
/* Program which finds whether the entered number is even or odd */


import java.io.*;
import java.net.*;


public class udpServerEO {
   public static void main(String args[]) {
      try {
         DatagramSocket ds = new DatagramSocket(2000);


         byte b[] = new byte[1024];
         DatagramPacket dp = new DatagramPacket(b, b.length);


         ds.receive(dp);


         String str = new String(
               dp.getData(), 0, dp.getLength());


         System.out.println(str);


         int a = Integer.parseInt(str);
         String s;
```

```
            if (a % 2 == 0)
                s = "Number is even";
            else
                s = "Number is odd";

            byte b1[] = s.getBytes();

            DatagramPacket dp1 = new DatagramPacket(
                    b1, b1.length,
                    dp.getAddress(), dp.getPort());

            ds.send(dp1);
            ds.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

2. udpClientEO.java

```
/* Program which finds whether the entered number is even or odd */

import java.io.*;
import java.net.*;

public class udpClientEO {
    public static void main(String args[]) {
        try {
```

```
DatagramSocket ds = new DatagramSocket(1000);

BufferedReader br = new BufferedReader(
        new InputStreamReader(System.in));

System.out.println("Enter a number : ");
String num = br.readLine();

byte b[] = num.getBytes();

DatagramPacket dp = new DatagramPacket(
        b, b.length,
        InetAddress.getLocalHost(), 2000);

ds.send(dp);

byte b1[] = new byte[1024];

DatagramPacket dp1 = new DatagramPacket(
        b1, b1.length);

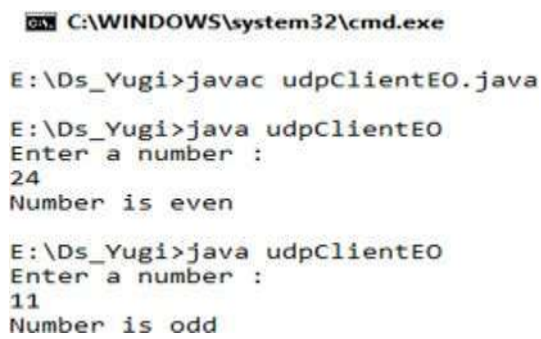
ds.receive(dp1);

String str = new String(
        dp1.getData(), 0, dp1.getLength());

System.out.println(str);
```

```
            br.close();
            ds.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

OUTPUT :





(B) AIM : A program to implement simple calculator operations like addition, subtraction, multiplication and division.

CODE :

1. RPCServer.java

```
import java.util.*;
import java.net.*;

class RPCServer {
   DatagramSocket ds;
   DatagramPacket dp;
   String str, methodName, result;
   int val1, val2;

   RPCServer() {
      try {
         ds = new DatagramSocket(1200);
         byte b[] = new byte[4096];

         while (true) {
            dp = new DatagramPacket(b, b.length);
            ds.receive(dp);

            str = new String(
                  dp.getData(), 0, dp.getLength());

            if (str.equalsIgnoreCase("q")) {
               break;
            }
```

```
StringTokenizer st = new StringTokenizer(str, " ");

methodName = st.nextToken();
val1 = Integer.parseInt(st.nextToken());
val2 = Integer.parseInt(st.nextToken());

System.out.println(str);

if (methodName.equalsIgnoreCase("add")) {
   result = "" + add(val1, val2);
} else if (methodName.equalsIgnoreCase("sub")) {
   result = "" + sub(val1, val2);
} else if (methodName.equalsIgnoreCase("mul")) {
   result = "" + mul(val1, val2);
} else if (methodName.equalsIgnoreCase("div")) {
   if (val2 != 0) {
      result = "" + div(val1, val2);
   } else {
      result = "Division by zero is not allowed";
   }
} else {
   result = "Invalid method name";
}

byte b1[] = result.getBytes();

DatagramPacket dp1 = new DatagramPacket(
```

```
                b1, b1.length,
                dp.getAddress(), 1300);

            System.out.println("Result : " + result + "\n");

            ds.send(dp1);
        }

        ds.close();
    } catch (Exception e) {
        e.printStackTrace();
    }
}

public int add(int val1, int val2) {
    return val1 + val2;
}

public int sub(int val1, int val2) {
    return val1 - val2;
}

public int mul(int val1, int val2) {
    return val1 * val2;
}

public int div(int val1, int val2) {
    return val1 / val2;
```

```
    }

    public static void main(String args[]) {
        new RPCServer();
    }
}
```

2. RPCClient.java

```
import java.io.*;
import java.net.*;

class RPCClient {
  RPCClient() {
        try {
            InetAddress ia = InetAddress.getLocalHost();

            DatagramSocket ds = new DatagramSocket();
            DatagramSocket ds1 = new DatagramSocket(1300);

            System.out.println("\nRPC Client\n");
            System.out.println(
                    "Enter method name and parameters like: add 3 4");
            System.out.println("Type q to quit.\n");

            BufferedReader br = new BufferedReader(
                    new InputStreamReader(System.in));

            while (true) {
                String str = br.readLine();
```

```
    if (str.equalsIgnoreCase("q")) {
       byte b[] = str.getBytes();
       DatagramPacket dp = new DatagramPacket(
             b, b.length, ia, 1200);
       ds.send(dp);
       break;
    }

    byte b[] = str.getBytes();

    DatagramPacket dp = new DatagramPacket(
          b, b.length, ia, 1200);

    ds.send(dp);

    byte b1[] = new byte[4096];
    DatagramPacket dp1 = new DatagramPacket(
          b1, b1.length);

    ds1.receive(dp1);

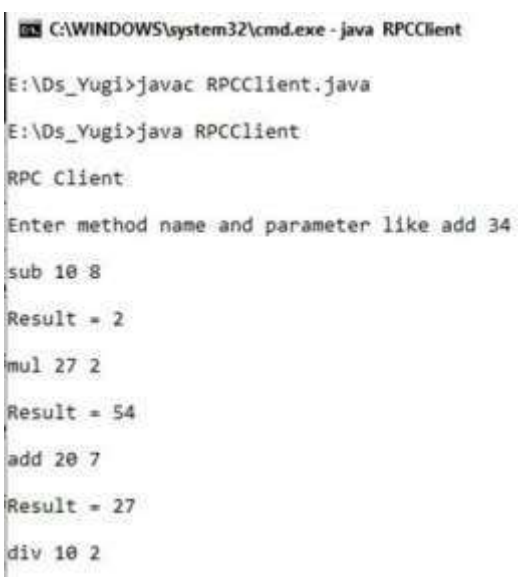
    String s = new String(
          dp1.getData(), 0, dp1.getLength());

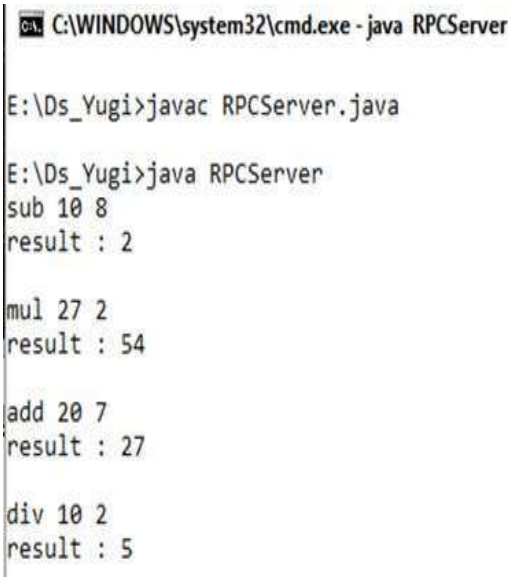
    System.out.println("\nResult = " + s + "\n");
  }
```

```
            ds.close();
            ds1.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    public static void main(String args[]) {
        new RPCClient();
    }
}
```

OUTPUT :





Practical 3

AIM : Multicast Socket example.

CODE :

1. BroadcastServer.java

```
import java.net.*;
import java.io.*;
import java.util.*;

public class BroadcastServer {
   public static final int PORT = 1234;

   public static void main(String args[]) throws Exception {
      MulticastSocket socket;
      DatagramPacket packet;
      InetAddress address;

      // Set the multicast address
      address = InetAddress.getByName("239.1.2.3");
      socket = new MulticastSocket();

      // Join the multicast group
      socket.joinGroup(address);

      byte[] data;

      for (;;) {
         Thread.sleep(10000);
```

```
            System.out.println("Sending");

            String str = "This is Neha Calling.....";
            data = str.getBytes();

            packet = new DatagramPacket(
                  data, data.length, address, PORT);

            // Send the packet
            socket.send(packet);
         }
      }
}
```

2. BroadcastClient.java

```
import java.net.*;
import java.io.*;

public class BroadcastClient {
   public static final int PORT = 1234;

   public static void main(String args[]) throws Exception {
      MulticastSocket socket;
      DatagramPacket packet;
      InetAddress address;

      // Set the multicast address
      address = InetAddress.getByName("239.1.2.3");
      socket = new MulticastSocket(PORT);

      // Join the multicast group
      socket.joinGroup(address);

      byte[] data = new byte[1024];

      packet = new DatagramPacket(data, data.length);

      for (;;) {
         // Receive the packets
         socket.receive(packet);

          String str = new String(
```

```
                packet.getData(),
                0,
                packet.getLength());


            System.out.println(
                "Message received from "
                + packet.getAddress());


            System.out.println("Message is : " + str);
        }
    }
}
```

OUTPUT :



Practical 4

AIM : RMI based application program to display current date and time.

CODE :

1. InterDate.java

```
import java.rmi.*;


public interface InterDate extends Remote {
   public String display() throws Exception;
}
```

2. ServerDate.java

```
import java.rmi.*;
import java.rmi.server.*;
import java.util.*;


public class ServerDate extends UnicastRemoteObject
       implements InterDate {


   public ServerDate() throws Exception {
      super();
   }


   public String display() throws Exception {
      String str = "";
      Date d = new Date();
      str = d.toString();
      return str;
   }
```

```
    public static void main(String args[]) throws Exception {
        ServerDate s1 = new ServerDate();

        Naming.rebind("DS", s1);

        System.out.println(
                "Object registered. ..... ");
    }
}
```

3. ClientDate.java

```
import java.rmi.*;
import java.io.*;

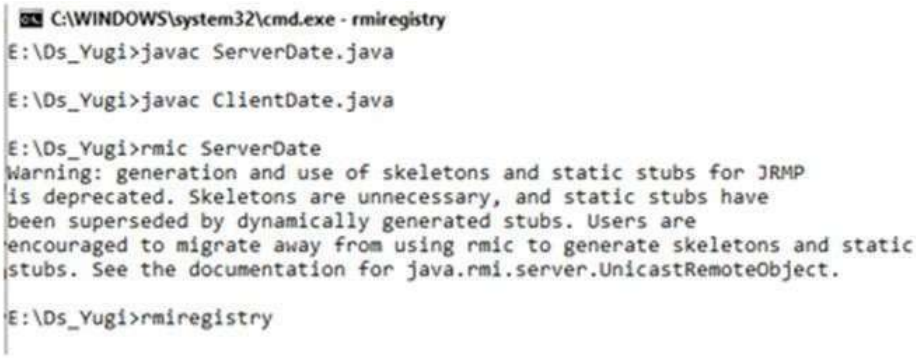
public class ClientDate {
    public static void main(String args[]) throws Exception {
        String s1;

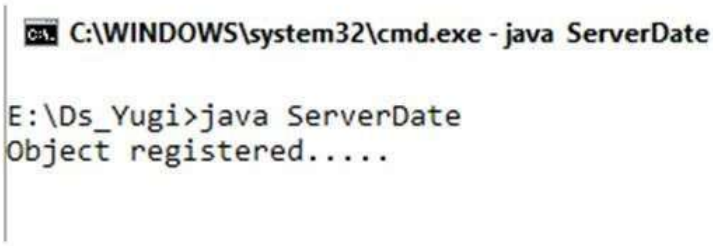
        InterDate h1 = (InterDate) Naming.lookup("rmi://localhost/DS");

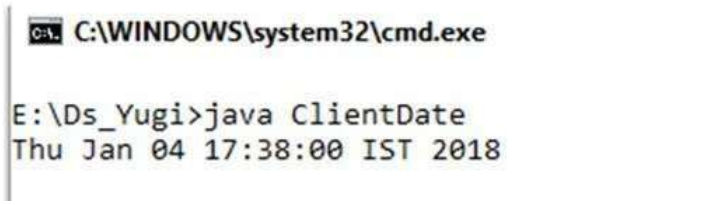
        s1 = h1.display();

        System.out.println(s1);
    }
}
```

OUTPUT :


C:\WINDOWS\system32\cmd.exe - rmiregistry
E:\Ds_Yugi>javac ServerDate.java
E:\Ds_Yugi>javac ClientDate.java
E:\Ds_Yugi>rmic ServerDate
Warning: generation and use of skeletons and static stubs for JRMP
is deprecated. Skeletons are unnecessary, and static stubs have
been superseded by dynamically generated stubs. Users are
encouraged to migrate away from using rmic to generate skeletons and static
stubs. See the documentation for java.rmi.server.UnicastRemoteObject.
E:\Ds_Yugi>rmiregistry


C:\WINDOWS\system32\cmd.exe - java ServerDate
E:\Ds_Yugi>java ServerDate
Object registered.....


C:\WINDOWS\system32\cmd.exe
E:\Ds_Yugi>java ClientDate
Thu Jan 04 17:38:00 IST 2018

Practical 5

AIM : Implement Xen virtualization and manage with Xen Center.

- Install **XenServer** in VMware Workstation and select Guest operating system as Linux.

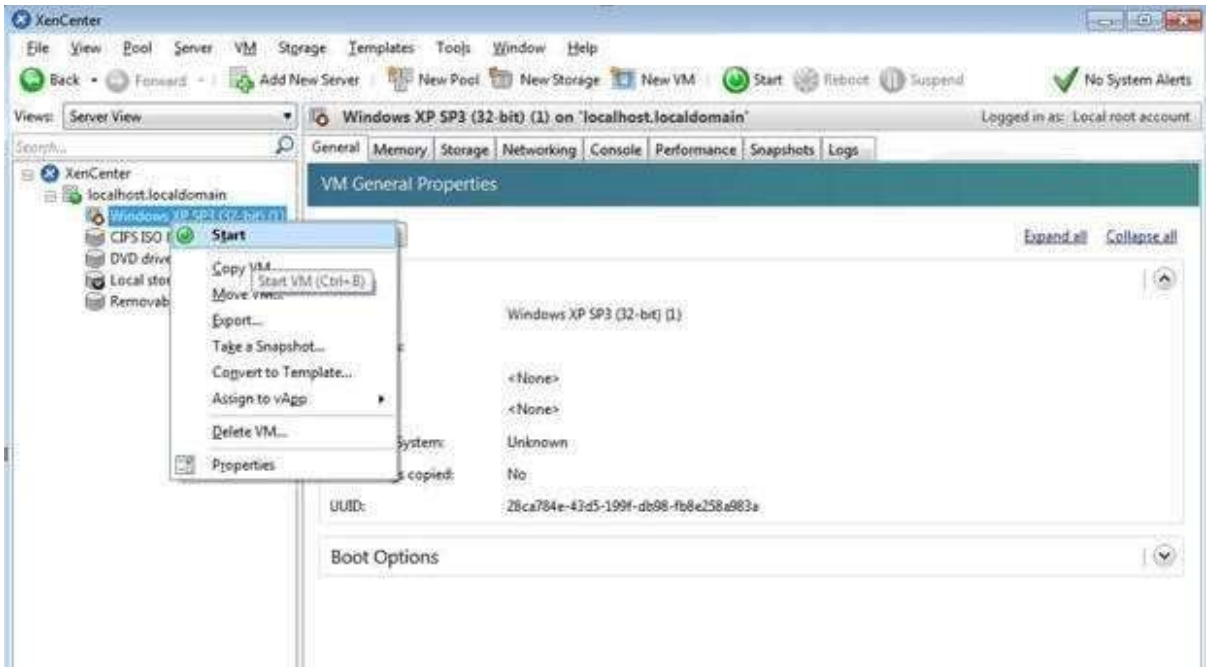


Note IP Address – " 192.168.124.137" ping it from command prompt.

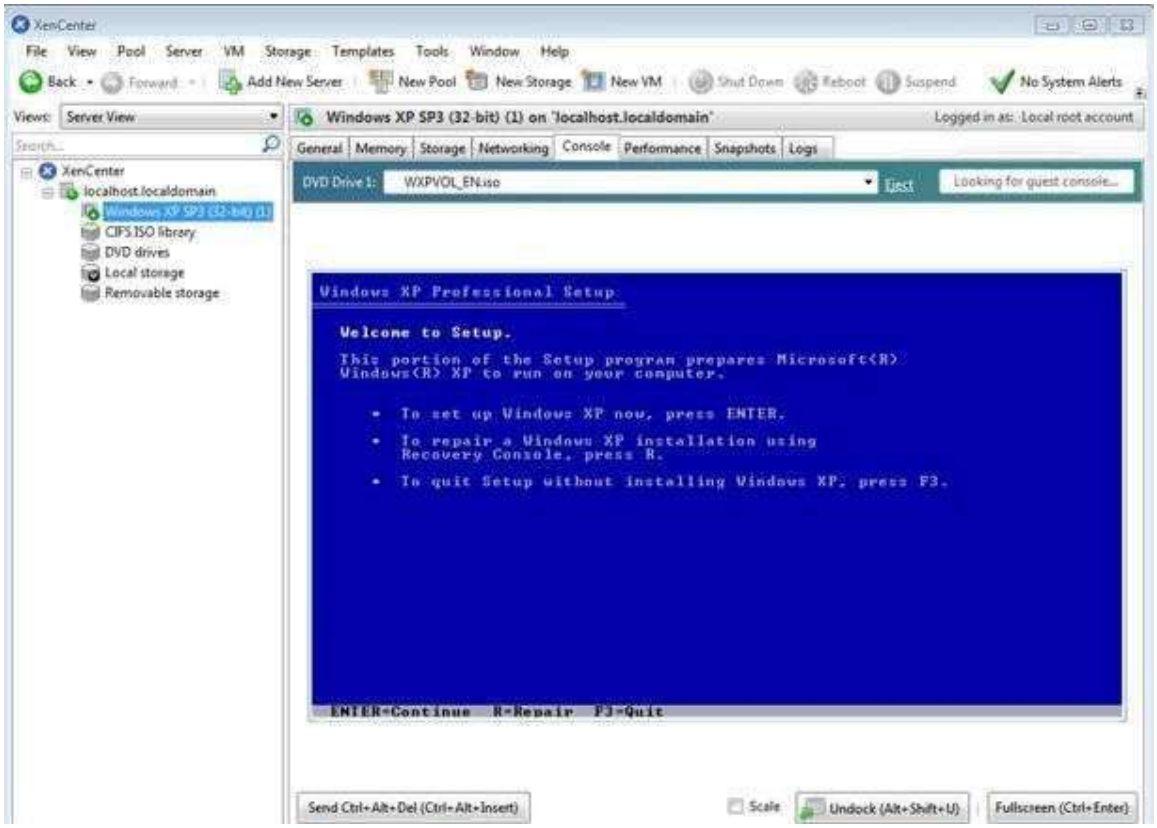
- Now Install Citrix App if not installed
- Now Open Citrix XenCenter – and Click and **Add Server.**



- Fill IP address copied from Installation and User name as “root” and Password as “root123” which we had given during installation and Click on Add.
- Then click on Ok
- Now Click on New Storage
- Select Window File Sharing (CIFS) and click on next
- Uncheck Auto generate option Click on Next.
- Provide the path of shared windows XP image and enter local pc credential , click on Finish
- Click on New VM – and Windows XP SP3
- Select ISO file and click on next –
- Click on Next –
- Uncheck – Start the new VM and click on create now
- Now Right click on Windows XP and Start -



Installation is successful and virtual node has been created if we get below Welcome screen of Windows XP machine.

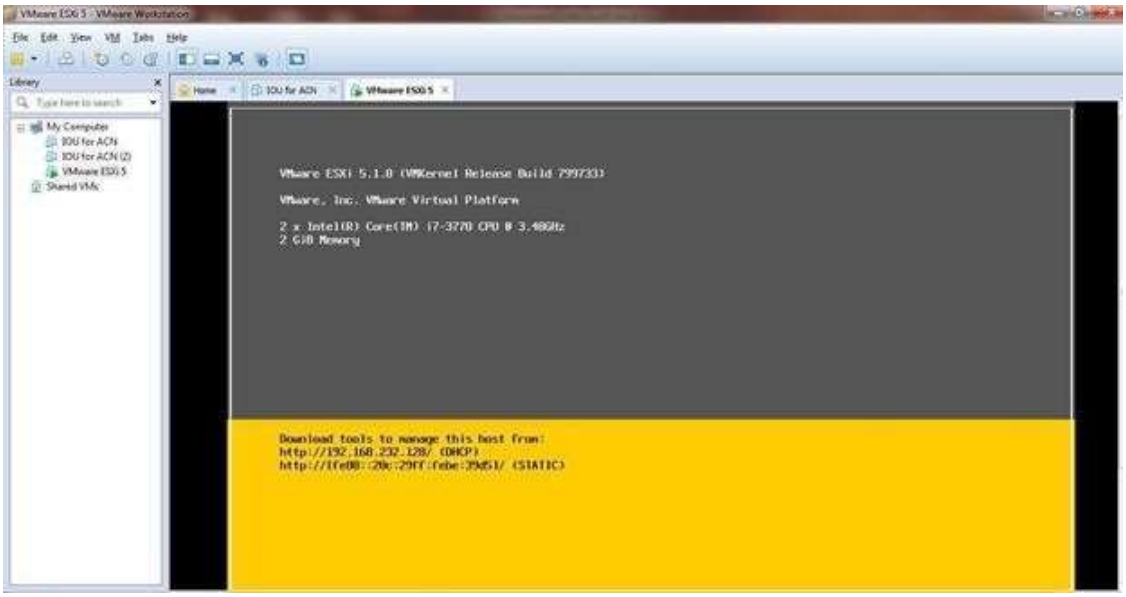


PRACTICAL 6

AIM : Implement virtualization using VMWare ESXi Server and managing with vCenter.

Steps:

Install ESXi iso in VMWare workstation.



Install VMware vSphere Client



In vSphere create new Virtual Machine. Install Windows XP iso file and open it.



Practical 7

AIM : Implement server cluster with Windows Server 2012.

Requirements: • Two Windows Server 2012 machines (Node1, Node2) in the same domain • One shared storage (iSCSI target) visible to both nodes • Static IP addresses on all machines

Step 1 : Open Server Manager → Add Roles and Features → Features → select Failover Clustering → Install (on both nodes).

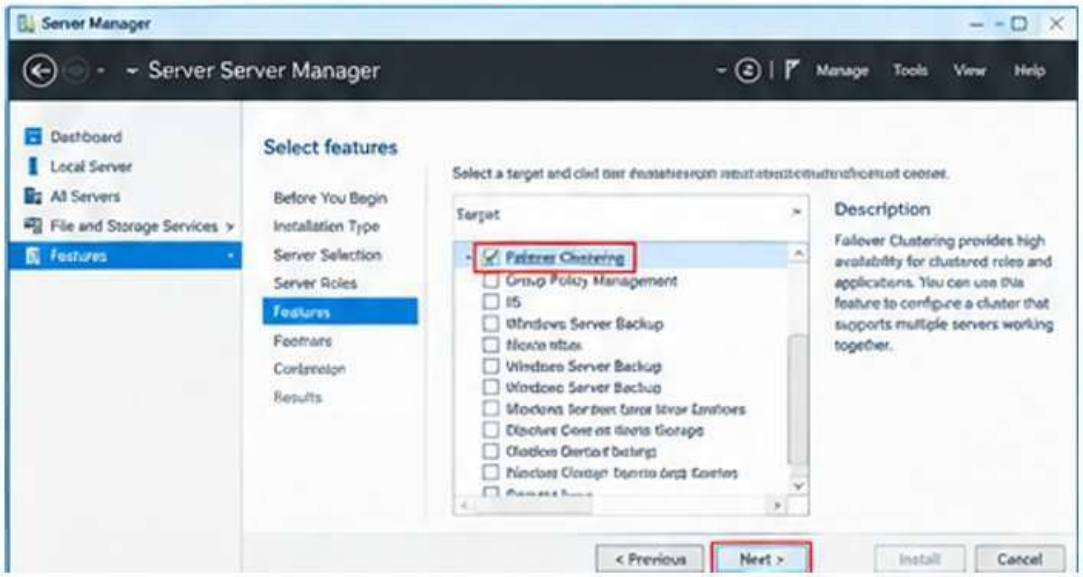
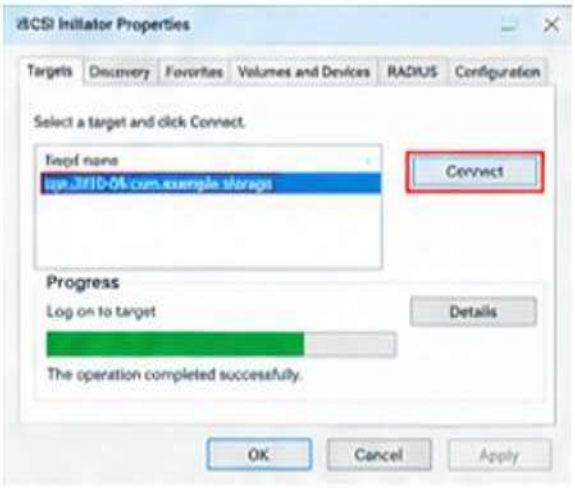


Figure 1: Failover Clustering installation

Step 2 : Create an iSCSI virtual disk and connect both nodes to it using iSCSI Initiator. Bring the disk online and format it (on Node1).



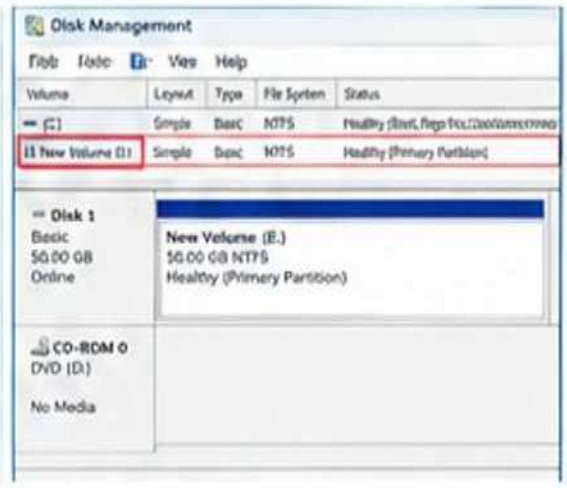


Figure 2: iSCSI initiator connected to shared disk, and the disk online and formatted on Node1

Step 3

Server Manager → Tools → Failover Cluster Manager → Validate Configuration → add Node1 and Node2 → run all tests.

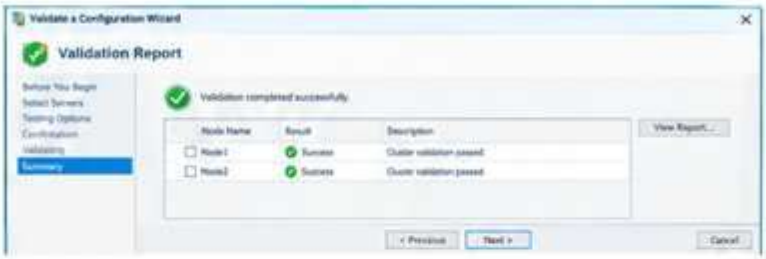


Figure 3: Validation report

Step 4

Click Create Cluster → add both nodes → enter cluster name and IP address → Finish.



Figure 4: Cluster creation summary

Step 5

Configure Role → select File Server (or any role) → choose the shared disk → Finish.



Figure 5: Role configured and running

Step 6

Test failover: shut down or stop the active node and verify the role moves to the other node.

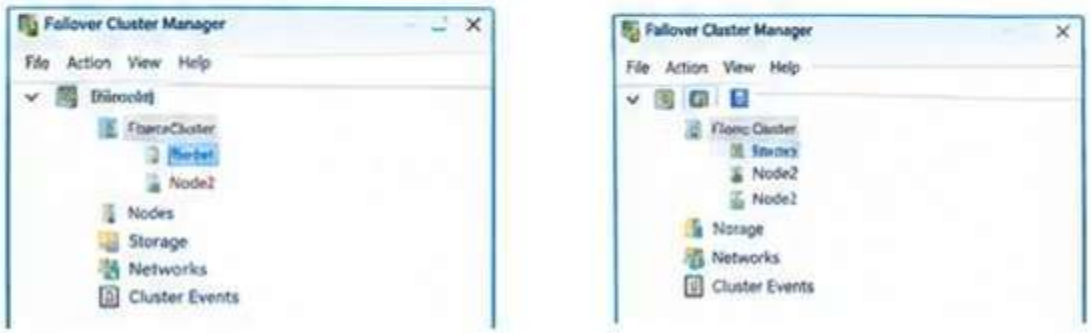


Figure 6: Role running on Node1 (active), then moved to Node2 after failover

Result: Failover cluster created and role successfully failed over between nodes.

Practical 8

AIM : Implement virtualization using Hyper-V.

First we have to uninstall vmware software if already installed on computer because the VMware Workstation installer does not support running on a Hyper-V virtual machine. after uninstalling vmware we can proceed to next step go to control panel and click on uninstall a program.

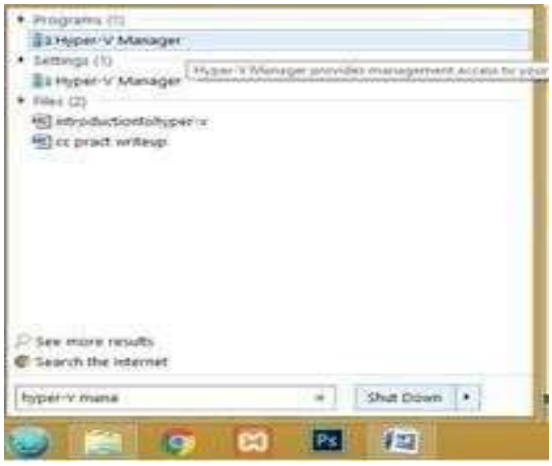


Click on Turn windows features on or off.

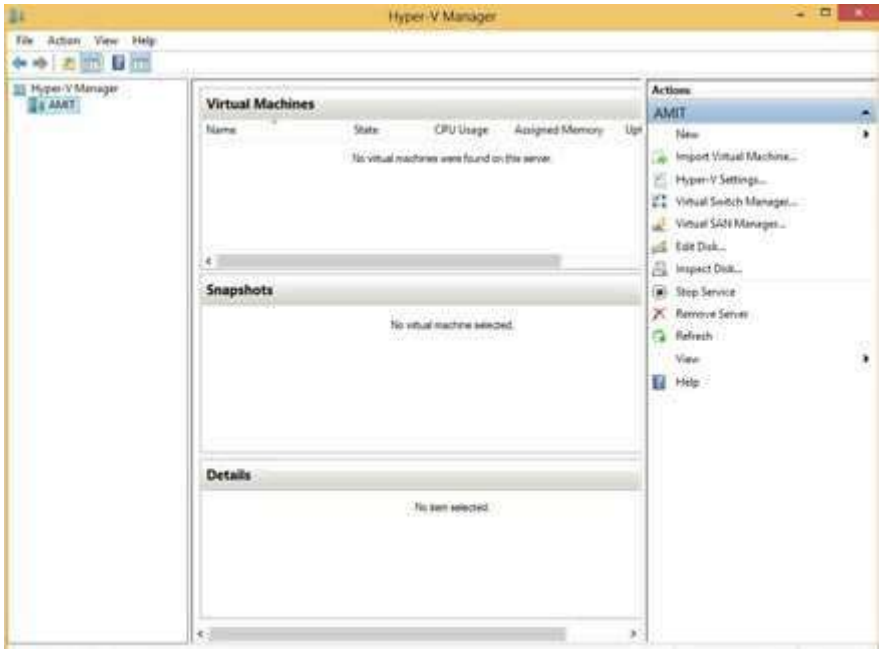
Now in windows features check on Hyper-V option.



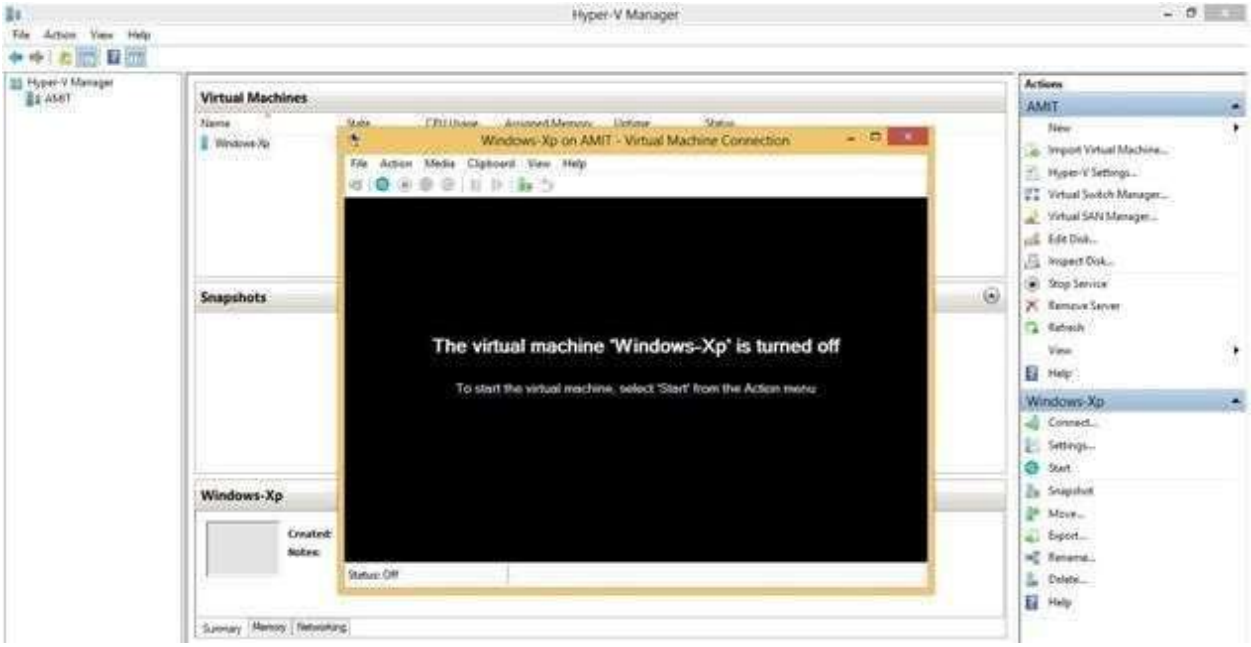
After Restart Search for hyper-v manager in search box and click on that.



for creating virtual machine first we have to create virtual switch click on virtual switch manager option.



Select External as a connection type and then click on create virtual switch. Create new Virtual switch and install windows XP .iso



and virtual machine will start.



PRACTICAL 9

AIM : Implement IaaS with Eucalyptus

Requirements:

• Clean CentOS/RHEL machine with virtualization enabled

• Static IP and internet access

• Free IP range for instances

Step 1 : Prepare the machine: set static IP, disable conflicting services, and update the system.

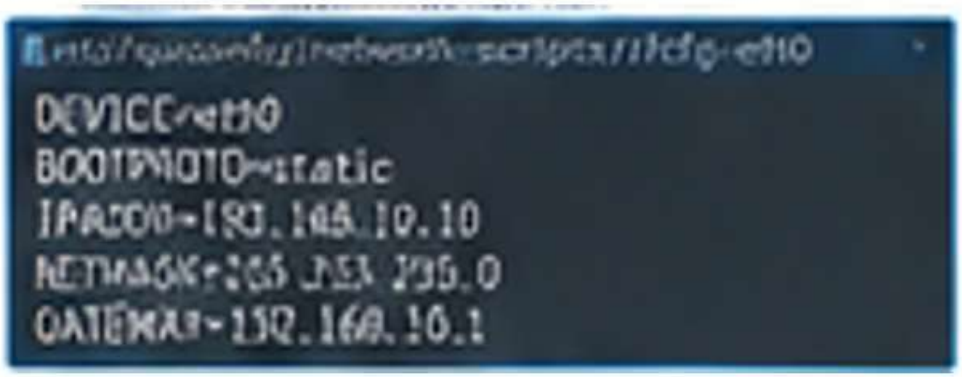


Figure 14: System network configuration

Step 2 : Run the Eucalyptus FastStart installer and answer the prompts (network mode, IP range).



Figure 15: FastStart installer running

Step 3 : Wait for installation to finish and note the console URL and login details.



Figure 16: Installation complete message

Step 4

Open the Eucalyptus Console in a browser and log in.



Figure 17: Eucalyptus dashboard

Step 5

Create a key pair and a security group (allow SSH, port 22).



Figure 18: Key pair and security group

Step 6

Images → select an image → Launch Instance → choose type, key pair and security group.



Figure 19: Launch instance wizard

Step 7

Verify the instance is Running and connect to it using SSH.



Figure 20: Running instance and SSH connection

Result: IaaS cloud set up with Eucalyptus and an instance launched successfully.

PRACTICAL 10

AIM : Implement OpenNebula

STEP 1: Update the system and note the IP address.

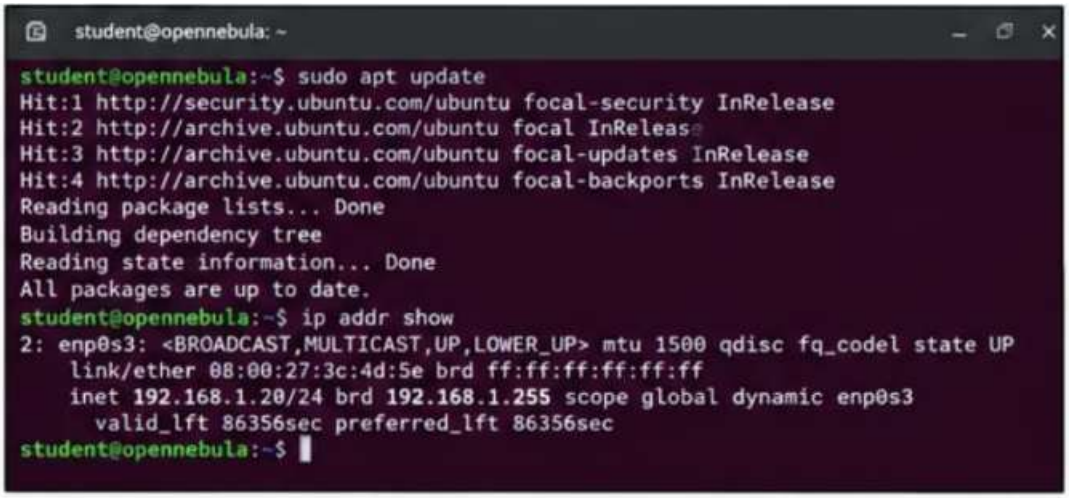


Figure 1: System update and IP address

STEP 2 : Download and run the miniONE installer.



Figure 2: miniONE installer running

STEP 3 : Wait for installation to finish and note the Sunstone URL , username(oneadmin) and password.



Figure 3: Installation complete message with login details

STEP 4 :

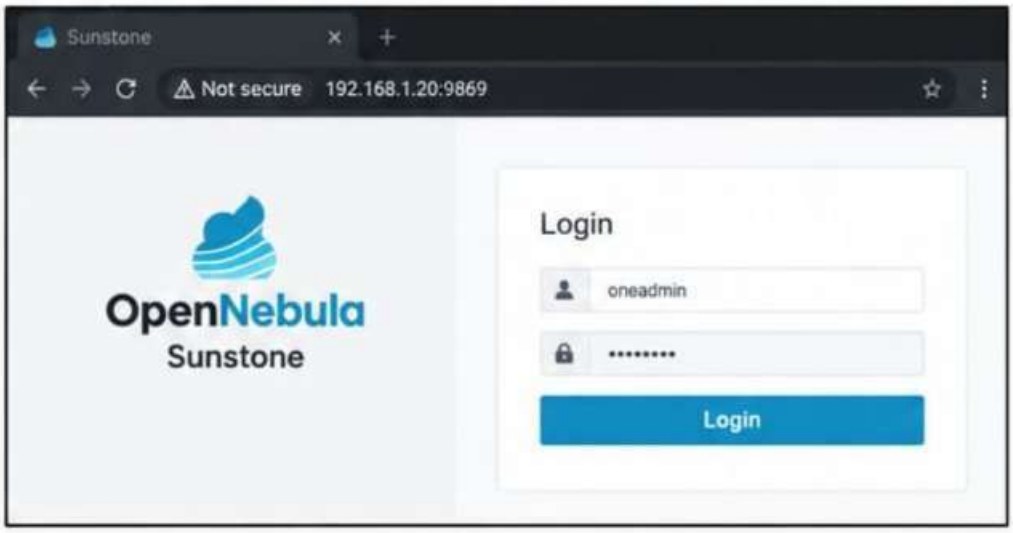


Figure 4: Sunstone login page

STEP 5 : Infrastrusture -> Hosts(showing the available hosts)



Figure 5: Host listed in Sunstone dashboard

